

Inverter (LT3462A): Working Document

Digital Audio Player • Rev 7

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1. Summary	1
2. Design Requirements	1
3. Pin Configuration	2
4. External Components	2
5. Equations and Results	2
6. Important Values	3
7. Open Problems	3
8. Revision Log	3
9. Sources	3

1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LT3462AES6 (AL-PN #1-089), U201
Package	6-lead TSOT-23 (S6)
Function	2.7 MHz current-mode inverting converter with integrated Schottky; +3V9 → −3.9 V (LT3093 input)

2. Design Requirements

Requirement	Value
Input	+3V9 from LTC3440, 3.827 to 3.981 V (3.74 V minimum during a load step)
Output	−3.9 V, regulated
Load	LT3093 only
Output current	≈ 27 mA quiescent; ≈ 98 mA peak; 126.6 mA LT3093 limit
Topology	Dual-inductor inverting (Ćuk)
Switching frequency	2.7 MHz fixed (2.0 to 3.5 MHz), no Burst Mode
Enable	Follows +3V9 (SDREF internal pull-up)
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	SW	L1 (L201), C1 (+)
2	GND	GND
3	FB	R1 / R2 midpoint, C4 to -3V9
4	SDREF	R2 to FB only; no GPIO (absolute maximum 2.5 V), no other load
5	D	C1 (-), L2 (L202)
6	VIN	+3V9, C_IN

Source: [1], [2]

4. External Components

Ref	Value / Rating	MPN	AL-PN
L1, L2 (L201, L202)	22 μ H $\pm$ 20 %, 400 mA I_{sat} , 810 m Ω max, 1210	Murata LQW32FT220M2HL	#1-090
C1, flying (C201)	2.2 μ F 25 V X7R 0603	Murata GRM188Z71E225KE43D	#1-070
C_IN, C_OUT (C211, C206)	10 μ F 10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R1 (R201)	309 k Ω 0.1 % 0402	YAGEO RE0402BRE07309KL	#1-091
R2 (R204)	100 k Ω 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131
C4 (C205)	22 pF 5 % COG 0402	Murata GRM1555C1H220JA01D	#1-093

Source: [2]

5. Equations and Results

$$V_{OUT} = -1.265 V \cdot R1/R2 = -1.265 \times 309/100 = -3.909 V (-3.816 \text{ to } -3.971 V) \quad (1)$$

$$I_{R2} = 1.265 V/100 k\Omega = 12.7 \mu A \text{ (SDREF sources up to } 80 \mu A) \quad (2)$$

$$D = \frac{|V_{OUT}| + V_D}{V_{IN} - V_{CE} + |V_{OUT}| + V_D} = \frac{4.709}{3.634 + 4.709} = 56.4 \% \quad (3)$$

$$\Delta I_L = \frac{V_{IN} D}{fL} = \frac{3.904 \times 0.564}{2.7 MHz \times 22 \mu H} = 37 mA_{pp} \quad (4)$$

$$I_{SW, pk} = I_{L1} + I_{L2} + \Delta I_L = 151 + 98 + 37 = 286 mA \quad (5)$$

$$I_P \leq V_{IN} \sqrt{C1/L1} = 3.9 V \sqrt{2.2 \mu F/22 \mu H} = 1.23 A \text{ (Schottky surge } 1.5 A) \quad (6)$$

Source: [1]. The inrush figure is an undamped upper bound with nominal C1; DC-bias derating at $\approx 7.8 V$ lowers it.

Item (simulated, 3.9 V in, C1 = 1 μ F)	98 mA load	27 mA load
V_OUT	-3.929 V	-3.938 V
Efficiency	67.7 %	71.8 %

Item (simulated, 3.9 V in, C1 = 1 μ F)	98 mA load	27 mA load
Input current (LTC3440 load)	141 mA avg	35 mA avg
Switch peak	0.282 A	0.100 A
Output ripple	0.52 mV pp	0.78 mV pp

Load steps 27 $\leftrightarrow$ 98 mA with C4 = 22 pF: +26 / -18 mV, single overshoot. Simulations used C1 = 1 μ F; the author accepts the 2.2 μ F part without re-simulation.

6. Important Values

Item	Value
Switch current limit	300 mA min / 420 mA typ
Switch V_CESAT	270 mV typ at 250 mA
Schottky forward drop	800 mV typ / 1100 mV max at 250 mA
Maximum duty cycle	77 %
SDREF reference	1.265 V (1.245 to 1.285 V)
Supply current	2.9 mA typ not switching; < 10 μ A shutdown
Switch peak at 98 mA	0.282 A simulated vs 300 mA min (6 % margin)
IC dissipation at 98 mA	$\approx$ 160 mW $\rightarrow$ $\approx$ 24 $^{\circ}$ C rise (θ_{JA} 150 $^{\circ}$ C/W)

Source: [1]

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0 to 5	2026-09-23	Initial issue; L1/L2, C1, C4 selection; efficiency and load-step simulation; library parts.
6	2026-09-25	Reformatted.
7	2026-10-03	C1 2.2 μ F 25 V X7R 0603 (#1-070), accepted by the author; inrush bound 1.23 A. R2 0402 (#1-131). Designators added.

9. Sources

1. LT3462/LT3462A Datasheet (Analog Devices)
2. DAP schematic sheet DAP-Power-2, inverter (author, 2026-10-03)